

MELBOURNE REGIONAL OFFICE, AustraliaWMO: **948680**Lat: **37.8075S**Lon: **144.9700E**Elev: **32**StdP: **100.94**Time Zone: **10.00 (AUS)**Period: **94-14**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	5.7	-1.3	3.4	18.0	0.0	3.8	14.8	8.8	12.6	7.8	12.2	1.0	90	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.3	35.0	18.8	32.5	18.3	30.0	17.9	21.2	28.5	20.2	27.6	19.4	26.6	2.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	13.9	23.6	18.0	13.0	22.9	17.0	12.2	22.3	61.7	28.6	58.3	27.6	55.5	26.6	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.2	5.3	DB	2.4	39.8	1.0	2.4	1.7	41.5	1.1	42.9	0.6	44.2	-0.1	45.9	(4)
(5)			WB	1.8	23.0	0.8	1.2	1.2	23.8	0.7	24.5	0.2	25.2	-0.4	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	21.1	21.4	19.3	16.2	13.6	11.6	10.9	11.8	13.6	15.2	17.6	19.1		
	DBStd	4.63	3.91	3.71	3.32	2.78	2.30	1.93	1.80	2.13	2.66	3.06	3.48	3.59		
	HDD10.0	25	0	0	0	0	1	6	10	6	2	0	0	0		
	HDD18.3	1241	11	8	25	76	146	203	229	203	146	107	56	31		
	CDD10.0	2186	344	318	288	186	114	53	39	61	108	163	227	283		
	CDD18.3	361	96	93	56	12	1	0	0	0	3	11	33	56		
	CDH23.3	3338	916	826	466	82	2	0	0	0	17	123	342	563		
	CDH26.7	1430	450	370	181	16	0	0	0	0	2	32	123	257		
(14) Wind	WSAvg	1.9	2.0	1.9	1.8	1.7	1.7	1.8	1.9	2.0	2.1	2.0	2.0	1.9		
(15) Precipitation	PrecAvg	635	49	47	40	58	60	45	45	55	56	64	59	58		
	PrecMax	867	176	238	137	195	135	117	91	109	126	150	166	197		
	PrecMin	332	2	0	7	10	10	11	9	23	17	15	11	2		
	PrecStd	132	35	51	26	38	31	23	21	21	28	30	37	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	37.4	35.1	28.7	22.8	18.7	17.7	21.2	25.6	30.3	33.6	36.9		
		MCWB	20.0	19.6	18.6	16.2	14.3	12.8	11.3	13.2	13.9	15.6	17.9	18.0		
	2%	DB	35.1	34.0	31.1	25.6	20.6	17.0	16.0	18.9	22.5	26.7	30.1	32.8		
		MCWB	18.8	19.1	18.4	15.5	13.7	12.2	10.8	11.9	13.9	14.9	17.0	17.6		
	5%	DB	31.7	31.3	28.2	23.5	19.1	15.8	15.0	17.3	20.4	24.0	27.2	29.4		
		MCWB	18.8	18.8	17.6	15.2	13.2	11.4	10.3	11.0	12.9	14.3	16.5	17.3		
	10%	DB	28.2	28.3	25.5	21.5	17.8	14.9	14.1	15.8	18.7	21.4	24.5	26.0		
		MCWB	18.6	18.6	17.2	14.7	12.7	10.9	9.8	10.3	11.9	13.5	15.9	16.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	22.3	20.8	18.1	15.7	14.3	12.7	13.8	16.3	17.8	20.5	20.5		
		MCDB	27.8	29.1	28.4	24.9	19.0	16.9	15.9	18.5	21.8	24.3	27.4	28.0		
	2%	WB	21.4	21.2	19.7	17.1	14.9	12.9	11.7	12.6	14.7	16.4	18.8	19.3		
		MCDB	29.0	28.0	26.8	22.2	18.5	15.9	14.8	17.7	20.6	23.0	25.5	26.7		
	5%	WB	20.2	20.3	18.7	16.2	14.1	12.0	11.0	11.7	13.6	15.2	17.7	18.5		
		MCDB	28.5	27.1	25.3	21.3	17.9	15.0	13.9	16.0	19.3	21.8	24.8	25.7		
	10%	WB	19.2	19.5	17.9	15.3	13.3	11.3	10.4	10.9	12.5	14.2	16.6	17.6		
		MCDB	26.6	26.2	24.3	20.3	17.0	14.3	13.3	14.8	17.7	20.0	23.1	24.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.3	8.6	8.1	6.5	5.6	5.7	6.8	7.9	8.7	9.1	9.5		
		MCDBR	16.7	15.4	14.2	11.9	8.5	6.8	6.7	9.1	10.8	14.0	15.1	16.6		
		MCWBR	4.8	4.5	4.5	4.9	4.2	3.7	3.6	4.3	5.0	5.5	5.3	5.0		
		MCDBR	12.3	11.4	11.0	9.5	6.9	5.6	5.7	8.0	9.5	11.1	11.6	12.4		
	5% WB	MCWBR	5.1	4.6	4.5	4.6	4.0	3.4	3.4	4.2	4.8	5.2	5.0	5.1		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.357	0.332	0.332	0.314	0.303	0.304	0.316	0.336	0.338	0.349	0.343		
	taud		2.507	2.501	2.569	2.540	2.576	2.593	2.582	2.537	2.475	2.490	2.487	2.523		
	Ebn at Noon		981	952	937	873	831	812	834	878	916	960	977	994		
	Edh at Noon		112	108	93	84	70	64	69	82	100	108	114	111		
(44) All-Sky Solar Radiation	RadAvg		7.15	6.24	4.98	3.45	2.28	1.77	1.98	2.80	3.99	5.34	6.29	7.05		
	RadStd		0.45	0.33	0.28	0.20	0.14	0.14	0.11	0.19	0.29	0.43	0.38	0.40		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.56	+0.40	N/A	N/A	+0.44	N/A	-14	-153	+194	+70
Regional (40 neighbors)		+0.50	N/A	N/A	+0.85	N/A	N/A	-17	-127	+170	+58

Nomenclature: See separate page